Himasriya Engu

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Oviedo, FL 32765

Research Interests

Medical Image Analysis, Computer Vision, Deep Learning, AI for Healthcare, Multi-class Image Classification, CNN Architectures

Education

Master of Science in Computer Science Aug 2025 – Present
University of Central Florida
GPA: 3.92 / 4.0
Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Advanced Algorithms

Bachelor of Technology in Computer Science and Engineering Aug 2021 – Apr 2025
Malla Reddy Engineering College for Women
First Class with Distinction — GPA: 3.56 / 4.0

Publications

Eye Deep-Net: a deep neural network-based multi-class retinal disease diagnostic
Himasriya Engu, Anil P Jawalkar, Navya D.
Turkish Journal of Computer and Mathematics Education (TURCOMAT), 15(3), Sep 2024.
DOI: 10.61841/turcomat.v15i3.14780

Research Experience

Eye Deep-Net: Multi-Class Retinal Disease Diagnosis (Deep Learning, Medical Imaging) 2024

- Designed and implemented CNN-based architectures for multi-class retinal disease classification (DR, MH, ODR, Normal).
- Conducted medical image preprocessing, including resizing, normalization, and data augmentation to improve model generalization.
- Trained and evaluated deep learning models using standard classification metrics and cross-validation techniques.
- Contributed to experimental design, implementation, and manuscript preparation for peer-reviewed publication.

Experience

EISystems May 2024 – Jun 2024
Data Science and Machine Learning Intern

- Developed a Titanic Survival Prediction model using supervised classification algorithms.
- Performed exploratory data analysis, feature engineering, and model evaluation to improve predictive performance.

DevTown Feb 2024 – Apr 2024
Full Stack Web Development Intern

- Built full-stack web applications using Flask, Django, and Express.js, integrating APIs and UI components.
- Deployed applications on AWS and managed databases using MySQL and MongoDB.

Selected Projects

Airfare Price Prediction Using Machine Learning Python

- Developed regression-based predictive models using historical airfare datasets and feature engineering.

Real-Time Vehicle Counting and Classification System Computer Vision

- Implemented a computer vision pipeline to detect, count, and classify vehicles from real-time video feeds.

Hybrid Movie Recommendation System Optimization Machine Learning

- Applied collaborative filtering and content-based filtering to generate personalized recommendations.

Technical Skills

Programming: Python, Java, C++, R, JavaScript, MATLAB
Deep Learning / Computer Vision: PyTorch, TensorFlow, CNNs, OpenCV
Machine Learning / Data: scikit-learn, Pandas, NumPy, Data Augmentation, Model Evaluation
Web: React, HTML, CSS, Flask, Django
Databases: MySQL, PostgreSQL, MongoDB
Tools: MLflow, Streamlit, LangChain, ChromaDB

Certifications

- ChatGPT Prompt Engineering for Developers – DeepLearning.AI
- Generative AI Learning Path – Google
- Oracle Certified Associate
- Python Certification – Coursera

Leadership & Activities

- Cultural Coordinator and Volunteer Head for technical events; active member of the college Dance Club.
- Participated in Medha Trail Blazer (2022, 2023, 2024), Idea Presentation (2024), and Smart India Hackathon (2025).